Calculus II

Solution to Assignment 8

Part I. Regular Problems

1.

$z = xy^3 - x^2y, \quad x = t^2 + 1, \quad y = t^2 - 1 \Rightarrow$

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt} = \left(y^3 - 2xy\right)(2t) + \left(3xy^2 - x^2\right)(2t) = 2t\left(y^3 - 2xy + 3xy^2 - x^2\right)$$

2.

$z = \tan^{-1}\left(x^2 + y^2\right), \quad x = s\ln t, \quad y = te^s \Rightarrow$

$$\begin{aligned}
\frac{\partial z}{\partial s} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s} = \frac{2x}{1 + (x^2 + y^2)^2} \cdot \ln t + \frac{2y}{1 + (x^2 + y^2)^2} \cdot te^s \\
&= \frac{2}{1 + (x^2 + y^2)^2}\left(x\ln t + yte^s\right) \\
\frac{\partial z}{\partial t} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t} = \frac{2x}{1 + (x^2 + y^2)^2} \cdot \frac{s}{t} + \frac{2y}{1 + (x^2 + y^2)^2} \cdot e^s \\
&= \frac{2}{1 + (x^2 + y^2)^2}\left(\frac{xs}{t} + ye^s\right)
\end{aligned}$$

3.

$$f(x, y, z) = y^2 e^{xyz}$$

(a)

$$\begin{aligned}
\nabla f(x, y, z) &= (f_x(x, y, z), f_y(x, y, z), f_z(x, y, z)) = \left(y^2 e^{xyz}(yz), y^2 \cdot e^{xyz}(xz) + e^{xyz} \cdot 2y, y^2 e^{xyz}(xy)\right) \\
&= \left(y^3 z e^{xyz}, \left(xy^2 z + 2y\right)e^{xyz}, xy^3 e^{xyz}\right)
\end{aligned}$$

(b) $\nabla f(0, 1, -1) = (-1, 2, 0)$

(c) $D_{\mathbf{u}}f(0,1,-1) = \nabla f(0,1,-1)\cdot\mathbf{u} = (-1,2,0)\cdot\left(\frac{3}{13},\frac{4}{13},\frac{12}{13}\right) = -\frac{3}{13}+\frac{8}{13}+0 = \frac{5}{13}$

4. (a) As shown in our lecture slides, $D_{\mathbf{u}}f = |\nabla f|\cos\theta$. Since the minimum value of $\cos\theta$ is -1 , occurring when $\theta = \pi$, the maximum rate of decrease of $D_{\mathbf{u}}f$ is $-|\nabla f|$ occurring when $\theta = \pi$; that is, when $\mathbf{u}$ is in the opposite direction of ∇f (assuming $\nabla f \neq \mathbf{0}$).

   (b) $f(x,y) = x^4y - x^2y^3 \Rightarrow \nabla f(x,y) = \left(4x^3y - 2xy^3, x^4 - 3x^2y^2\right)$, so f decreases fastest at the point $(2,-3)$ in the direction $-\nabla f(2,-3) = -(12,-92) = (-12,92)$. The maximum rate of decrease is

$$-|\nabla f(2,-3)| = -|(12,-92)| = -\sqrt{12^2+(-92)^2} = -\sqrt{8608} = -4\sqrt{538}$$

5. We have $f_x(x,y) = 1$ and $f_y(x,y) = 2y$.

   (a) At $(x,y) = (0,0)$, $f(0,0) = 0, f_x(0,0) = 1$ and $f_y(0,0) = 0$. Hence the equation of the tangent plane is $z = 0 + 1\cdot(x-0) + 0\cdot(y-0)$ or simply $z = x$.

   (b) At $(x,y) = (1,2)$, $f(1,2) = 5, f_x(1,2) = 1$ and $f_y(1,2) = 4$. Hence the equation of the tangent plane is $z = 5 + 1\cdot(x-1) + 4\cdot(y-2)$ or simply $z = x + 4y - 4$.

Part II. Group Problems

1. (a) $\frac{\mathrm{d}}{\mathrm{d}x}(f(x))^3 = 3(f(x))^2\cdot f'(x)$

   (b) $\frac{\partial}{\partial x}(y+x^2)^3 = 3(y+x^2)^2\cdot 2x = 6x(y+x^2)^2$

   (c) $\frac{\partial}{\partial x}g(y+x^2) = g'(y+x^2)\cdot 2x$

   (d) $\frac{\partial}{\partial y}(f(x,y))^3 = 3(f(x,y))^2\cdot f_y(x,y)$

2. (a) For $f(x,y) = x^2 - y^2$

$$f_{xx} = \frac{\partial}{\partial x}(2x) = 2, \quad f_{xy} = \frac{\partial}{\partial y}(2x) = 0, \quad f_{yy} = \frac{\partial}{\partial y}(-2y) = -2$$

   (b) For $f(x,y) = (x+y)^2$

$$f_{xx} = \frac{\partial}{\partial x}2(x+y) = 2, \quad f_{xy} = \frac{\partial}{\partial y}2(x+y) = 2, \quad f_{yy} = \frac{\partial}{\partial y}2(x+y) = 2$$

   (c) For $f(x,y) = e^{-x}\cos y$

$$\begin{aligned} f_{xx} &= \frac{\partial}{\partial x}(-e^{-x}\cos y) = e^{-x}\cos y, \\ f_{xy} &= \frac{\partial}{\partial y}(-e^{-x}\cos y) = e^{-x}\sin y, \\ f_{yy} &= \frac{\partial}{\partial y}(-e^{-x}\sin y) = -e^{-x}\cos y. \end{aligned}$$